

Supervised Learning: Regression Models and Performance Metrics

Question 1 : What is Simple Linear Regression (SLR)? Explain its purpose.

Answer:

Simple Linear Regression is a statistical method used to model the relationship between one independent variable (X) and one dependent variable (Y) using a straight line.

Purpose: To predict the value of Y based on X and understand how X influences Y.

Question 2: What are the key assumptions of Simple Linear Regression?

Answer:

1. Linearity – Relationship between X and Y is linear
2. Independence – Observations are independent
3. Homoscedasticity – Constant variance of errors
4. Normality – Errors are normally distributed
5. No multicollinearity (for multiple regression, not SLR)

Question 3: Write the mathematical equation for a simple linear regression model and explain each term.

Answer:

Equation:

$$Y = \beta_0 + \beta_1 X + \epsilon$$

Where:

- Y = Dependent variable
- X = Independent variable
- β_0 = Intercept (value of Y when $X = 0$)
- β_1 = Slope (change in Y for unit change in X)
- ϵ = Error term

Question 4: Provide a real-world example where simple linear regression can be applied.

Answer:

Predicting house price based on size (square feet).
Here, size = independent variable, price = dependent variable.

Question 5: What is the method of least squares in linear regression?

Answer:

It is a method used to find the best-fitting line by minimizing the sum of squared differences between actual values and predicted values.

Question 6: What is Logistic Regression? How does it differ from Linear Regression?

Answer:

Logistic Regression is used for classification problems where the output is categorical (e.g., 0 or 1).

Linear Regression	Logistic Regression
Predicts continuous values	Predicts probabilities/classes
Uses straight line	Uses sigmoid function
Output range: $(-\infty, \infty)$	Output range: $(0, 1)$

Question 7: Name and briefly describe three common evaluation metrics for regression models.

Answer:

- 1. Mean Absolute Error (MAE):** Average absolute difference between actual and predicted values
- 2. Mean Squared Error (MSE):** Average squared difference (penalizes large errors)
- 3. Root Mean Squared Error (RMSE):** Square root of MSE, easier to interpret

Question 8: What is the purpose of the R-squared metric in regression analysis?

Answer:

R-squared measures how well the model explains the variability of the dependent variable.
Range: 0 to 1
Higher value = better fit.

Question 9: Write Python code to fit a simple linear regression model using scikit-learn and print the slope and intercept.

(Include your Python code and output in the code box below.)

Answer:

```
import numpy as np
from sklearn.linear_model import LinearRegression

# Sample data
X = np.array([[1], [2], [3], [4], [5]])
Y = np.array([2, 4, 5, 4, 5])

# Model
model = LinearRegression()
model.fit(X, Y)

# Output
print("Slope:", model.coef_[0])
print("Intercept:", model.intercept_)
```

Output

```
Slope: 0.6
Intercept: 2.2
```

Question 10: How do you interpret the coefficients in a simple linear regression model?

Answer:

- ☐ **Intercept (β_0):** Value of Y when X = 0
 - ☐ **Slope (β_1):** Change in Y for each unit increase in X
- If slope is positive → Y increases
If slope is negative → Y decreases